

UNIT - I INTRODUCTION

Introduction to Robotics

Robotics is a branch of mechanical engineering, electrical engineering and Computer science engineering that deals with design, construction and application of robotics, as well as Computer system for processing. These technology deals with automated machine tool that takes place the humans in dangerous environment (a) Manufacturing process

(a)

Robotics is an applied engineering science that has been referred to as a combination of machine tool technology and computer science. It includes machine design, production theory, micro-electronics, computer programming and artificial intelligence.

Robot: The word robot has been derived from Czechoslovakia which means forced labour or forced worker. It has been derived from the word "ROBOTA" around 1920 in a "science fiction short story by carol Capek". It has taken another four years to use industries.

Definition of Robot: The official definition of an industrial robot is a re-programmable, multi-functional manipulator designed to move materials, parts, tools or special devices through variable programmed

motions for the performance of variety of tasks.

→ It is provided by the "Robotics Industry Association" (RIA), formerly the robotics institute of america (RIA).

Automation and Robotics Automation and Robotics are closely related to each other.

Automation Automation is a technology concerned with the application of mechanical, electronics, and computer based systems and used to operate and control the production in manufacturing industries.

Example Robotics, CAD/CAM, CIM, FMS

Reasons for implementation of automation system.

- (i) To increase the productivity
- To decrease the cost of labour
- To minimize the effect of shortage of labour.
- To improve the high quality of product
- To decrease the manufacturing lead time.

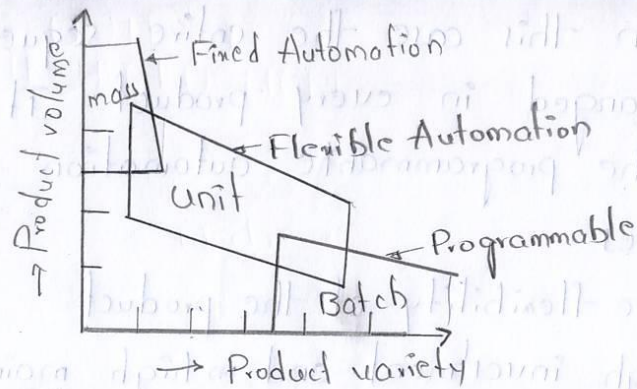
Safety of workers.

Types of Automation (i) Automated manufacturing Systems.

Fixed Automation

Programmable Automation

Flexible Automation



Fixed Automation: It is a system in which the sequence of processing operation is fixed by the equipment configuration each of the operation in the sequence is usually simple

Features

- 1) High productivity
- 2) Relatively flexible in product variety

Example: Automobile industry

Programmable Automation

It is a system in which the production equipment is designed with the capability to change the sequence of operations to accommodate different product configuration

Features

- 1) High Flexible compared to the fixed automation
- 2) Batch production is produced
- 3) High investment for general purpose

Example: CNC, NC etc

Flexible automation: In flexible automation system no. of variety products produced in same production

line in this case the entire sequence operation is changed in every product. It is extension for the programmable automation.

Features

- (1) More flexibility of the product
- (2) High investment and high maintainance also
- (3) Produce unit production

Example : FMC, chemical industry.

Of the three types of automation robotics coincides most closely with programmable automation. An industrial robot is a general purpose, programmable machine which process certain human like characteristics. The most typical human like characteristic of present day robot is their movable arms.

Advantages of automation

- (1) High production rates
- (2) Manufacturing lead time decreases
- (3) Storing capacity decreases
- (4) Human errors are eliminated
- (5) Labour cost is less

Disadvantages

- (1) Initial cost of raw material is very high
- (2) Maintainance cost is high
- (3) Requires high skilled operators
- (4) Indirect cost for research development & programming increases.

(5) Employment decreases.

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Need for using robots in industries

- (1) Robots can be built a performance capability is high compare to human beings like strength, size, speed, accuracy etc
- (2) Robots are better then humans to perform simply and difficult tasks with better quality.
- (3) Robots do not have limitations and -ve attributes of human workers such as need for rest, diversion of attention, absences.
- (4) Robots involved in poor working conditions.
- (5) Improved working conditions and reduced risks.

CAD/CAM and Robotics

It is a technology concerned with the use of digital computers to perform certain functions in design and production.

CAD: Computer aided design is defined as the use of computer system to assist in the creation, modification analysis (or) optimisation of design.

CAM: Computer aided manufacturing can be defined as the use of computer system to plan, manage and control the operation of a manufacturing through direct (or) indirect computer interface.

Robotics in Manufacturing

Robot work cells can be organised into various arrangement (or) layouts. These layouts can be classified into 3 type.

1) Robot centered cell

2) Inline robot cell

3) Mobile robot cell

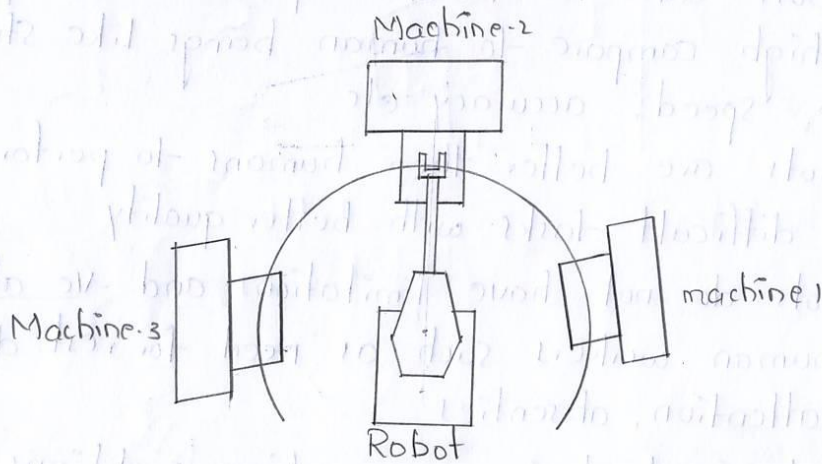
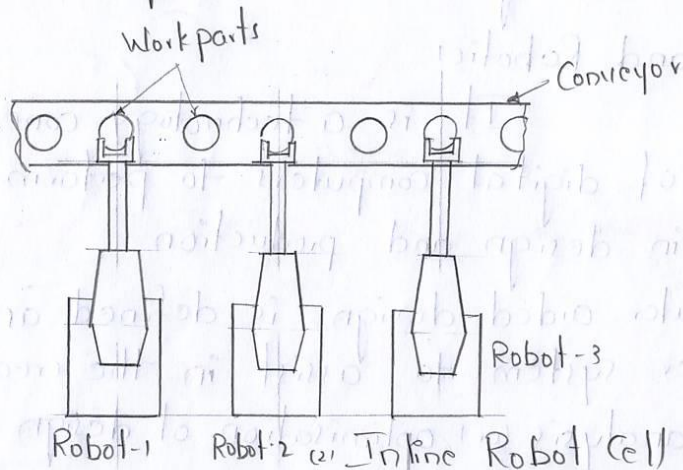
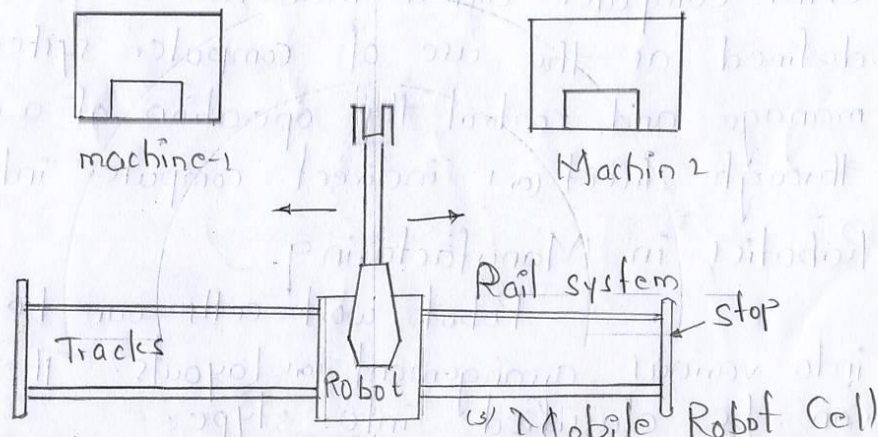


Figure shows 1) Robot Centered Cell



2) Inline Robot Cell



3) Mobile Robot Cell

Robot centered cell: In the robot centered cell the robot is located at the approximate centre of the cell and the equipment is arranged in a partial circle around it as shown in above figure. The most elementary case is where one robot performs a single operation, either servicing a single production machine or a single production operation. Here the main application of robot is to loading and loading.

Inline Robot cell: The robot is located along a moving conveyor or other handling system and performs a task on the product as it travels past on the conveyor. A common example of this cell is found in carbody assembly plants in automobile industries. Robots are positioned along the assembly line to spot weld the car body frames and panels.

Mobile robot cell: In which the robot is capable of moving the various pieces of equipment within the cell this is generally accomplished by mounting the robots on mobile base which can be transported on a rail system. The rail system used in robot cells are either tracks on floor of the plant or Over head rail system.

Present applications of robot in manufacturing industry

→ Material transfer

- Machine loading and unloading
- Processing operations like spot welding grinding drilling etc.
- Assembly tasks
- Inspection

Future Application

- Intelligence application
- Sensor capabilities
- Mechanical designs
- Universal gripper
- Robots in space
- Security guards
- Home need application
- Medical care and Hospital duties.

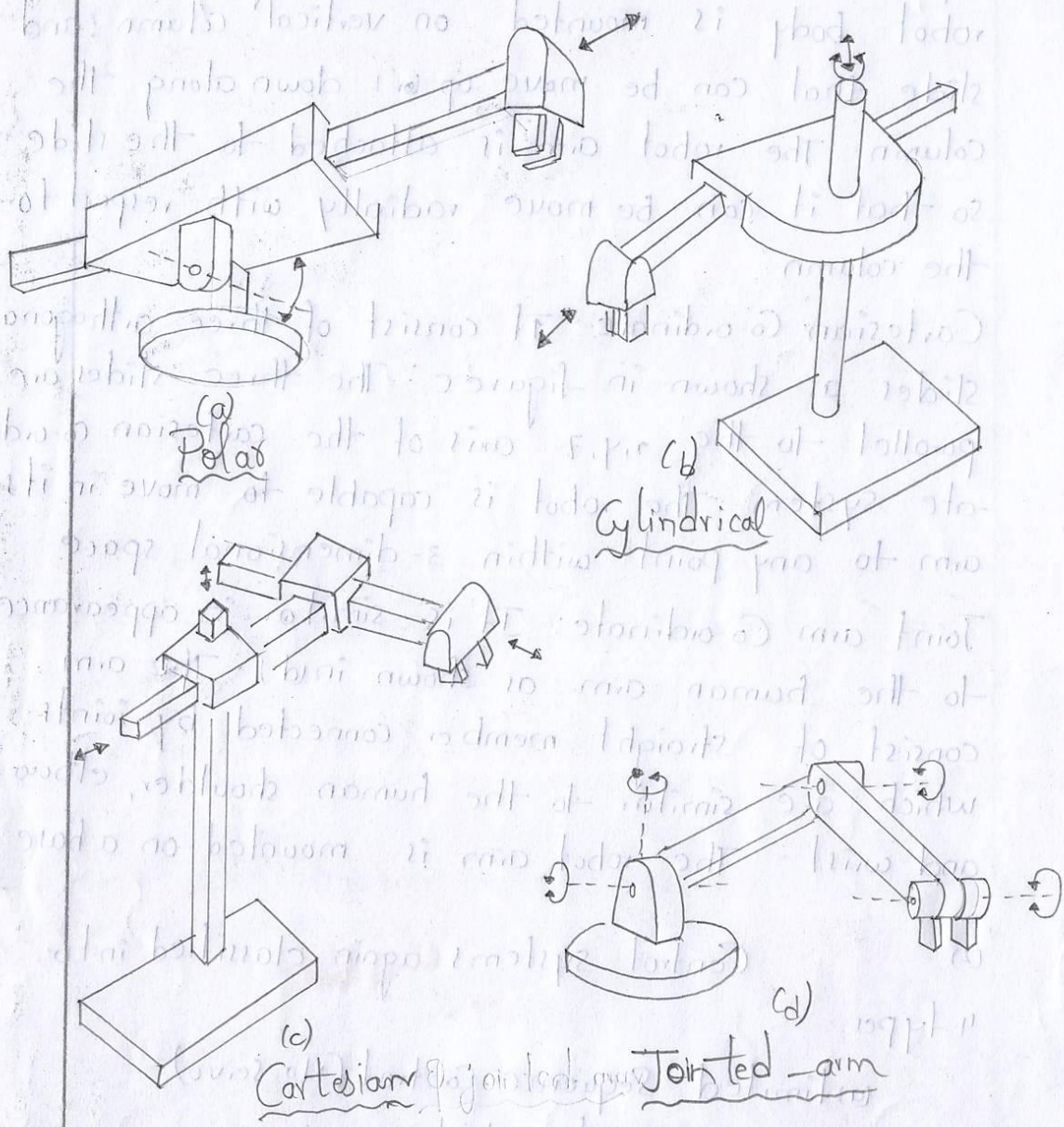
Classification of Robot.

Robots are classified into two main category

- 1) According to co-ordinate system
- 2) According to Control System.

According to Co-ordinate system

- a) Polar Configuration
- b) Cylindrical Configuration
- c) Cartesian Co-ordinate Configuration
- d) Jointed-arm Configuration



Polar-Configuration: This robot is also called as spherical co-ordinate robo because the work space within which it can move. Its arm is a partial sphere. The robot has a rotary base and a pivot that can be used to raise and lower a telescopic arm as shown in above figure.

Cylindrical Co-ordinate: In this configuration the robot body is mounted on vertical column and slide that can be move up or down along the column. The robot arm is attached to the slide so that it can be move radially with respect to the column.

Cartesian Co-ordinate: It consist of three orthogonal slides as shown in figure c. The three slides are parallel to the x, y, z axis of the cartesian co-ordinate system. The robot is capable to move in its arm to any point within 3-dimensional space.

Joint arm Co-ordinate: It is similar in appearance to the human arm as shown in (d). The arm consist of straight member connected by joints which are similar to the human shoulder, elbow and wrist. The robot arm is mounted on a base.

(2) Control systems again classified into 4 types

- (a) Limited Sequence Control (No servo)
- (b) Point to point control
- (c) Continuous path control
- (d) Intelligent robots

Limited Sequence Control: This type of robots do not give servo controlled to indicate relative position of the joint. Instead of controlled by

the switches and mechanical stops. Generally this type of robot involves simple motion. Example pick & place operations.

Point to point motion: This type of robots are capable of controlling velocity, acceleration and path of motion from the beginning to the end of the part. It uses complex control programs, PLCs (Programmable Logic Control) Computers and sensors to control the motion.

The point to point control motion robots are capable of performing motion cycle that consist of a series of desired point location. The robot is thought and recorded into these points are by controlled unit.

Example: Spot Welding.

Continuous path: The robot moves through a series of closely spaced points which describes the desired part. The individual points are defined by the control unit or programme.

Example: Spray painting, arc welding.

Intelligent Robot: This type of robots are not only programmable motion cycle. But also interact with its environment in a way that seems intelligent. It takes logical decisions based on sensor data receiving from the operator.

This type of robots usually programmed using an english like symbolic language not like computer programming language.

Precision of Movement (or) Parameters of Robot.

The following are

(i) Spatial Resolution

(ii) Accuracy

(iii) Repeatability

Spatial Resolution: Spatial Resolution of a robot is the smallest increment of movement into which the robot can divide its work value. The control resolution is determined by the robots position control system and its feed back measurement system.

Therefore the no. of increments 2^n where

$n = \text{No. of bites in the control memory.}$

Control Resolution = $\frac{\text{Total motion range}}{\text{No. of increments.}}$

Accuracy: It refers to robot ability to position it wrist at desired target position within the work volume.

Repeatability The robot ability to position it wrist at a point in space that had previously thought.

Robot with a single degree of freedom, 1 sliding joint with a full range of 1m. The robot's control memory has a 12 bit storage capacity. Determine the control resolution for the axis of motion.

Given

$$\text{Motion Range} = 1\text{m}$$

$$\text{Control Memory} = 12\text{ bit}$$

We know that

$$\text{Control increments} = 2^n$$

$$= 2^{12}$$

$$= 4096$$

$$\text{Control Resolution} = \frac{\text{Total motion range}}{\text{No. of increments}}$$

$$= \frac{1}{4096}$$

$$= 0.00024$$

Robot with a single degree of freedom, rotating
 joint with a full range of motion. The robot
 control necessary for a 1 bit mode is relatively
 determined. The control resolution for the control

motion
 range

Motion Range - 1 bit
 Control Memory - 1 bit

We know that

Control resolution = 1 bit

" = 1 bit

Mode = 1 bit

Control Resolution = Total motion range
 No. of increments

$$\frac{1}{2^{10}}$$

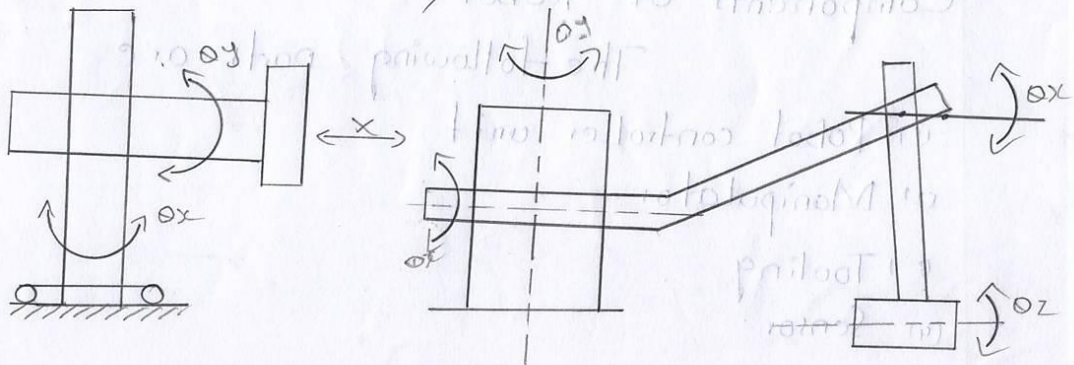
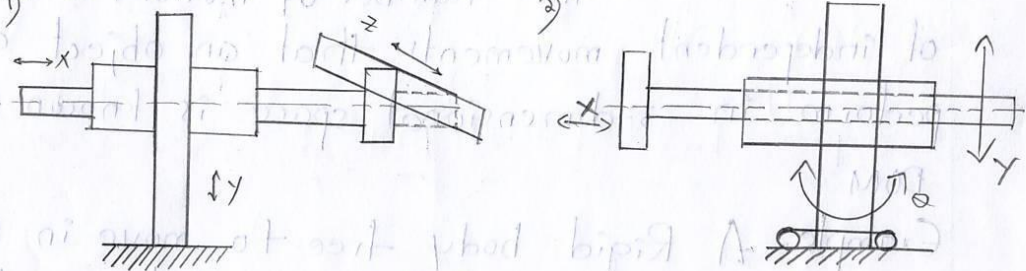
0.000976

COMPONENTS OF INDUSTRIAL ROBOTICS

Types of Co-ordinate systems, Types of arms

The following arms are:

- 1) Cartesian Robot
- 2) Cylindrical Robot
- 3) Polar Robot
- 4) Spherical Robot
- 5) Joint arm Robot
- 6) SCARA



A robotic arm is a type of mechanical arm usually programmable with similar function to a human arm. The arm may be the part of a more complex robot. The main purpose of arm is to hold the work parts with good accuracy and move from position to the next position.

It consists of no. of joints and no. of links. These links are connected through the manipulator. The main function of manipulator is to control the movements.

Degree of freedom (DOM)

The number of motions or Number of independent movements that an object can perform in 3-dimensional space is known as DOM.

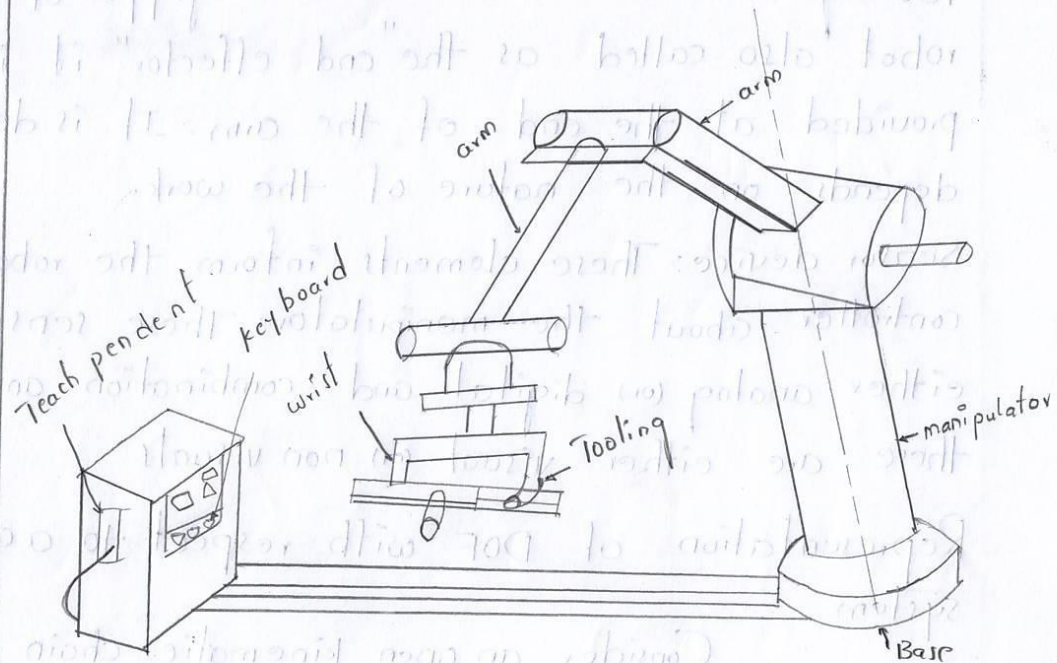
Example: A Rigid body free to move in space has got 6 degrees of freedom that is 3-translations or positions and 3-rotation or orientation.

Components of Robot.

The following parts are

- (1) Robot controller unit
- (2) Manipulator
- (3) Tooling
- (4) Sensor.

Main Components of Robot



Robot controller unit: The instructions to the robot to perform the desired task are into through the key board. A teachpendant is also provided for giving non textual commands input. The RCU converts the input programme to suitable signals which accurate the manipulator to perform particular task.

Manipulator: It consist of base arm and wrist similar to a human arm. It also improves power source either electric, hydraulic, pneumatic arm receiving signals from robot controller (RCU). The moment of manipulator can be relation to

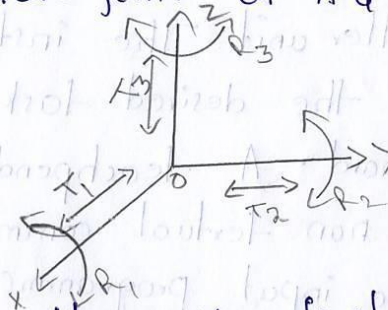
its co-ordinate system. which may be polar or cartesian etc.

Tooling: It is the hand or gripper of the robot also called as the "end effector" it is provided at the end of the arm. It is designed depends on the nature of the work.

Sensor device: These elements inform the robot controller about the manipulator. These sensors either analog or digital and combination and also these are either visual or non visual.

Representation of DOF with respect to a co-ordinate system.

Consider an open kinematic chain of two links with revolute joint at A & B or C as shown.



Here the first link is connected to the ground by joint A. Therefore link 1 can only rotate about joint A and it contributes only 1DOF link 2 can rotate with respect to contribute of joint 2. So other DOF. So the above figure has got 2DOF

→ The DOF is equal to the number of links in an open kinematic chain

→ Robot with 6 DOF is called as spatial manipulator.⁽⁶⁾
arm & body, wrist

→ Robot is more than 6 DOF is called as
redundant manipulator.

End effectors: At the end of the arm is known
as end effector. The main purpose of end effector
is to perform the specific task. It is attached to
the wrist. The end effector will have 3
motions, namely

(1) Roll → Rotational or swivel

(2) Pitch → Up or down

(3) Yaw → Right or left

Types of end effectors: There are two types of
end effectors.

(1) Gripper

(2) Tooling

Grippers: Grippers are used to hold either work
parts or tools. The type of material used for
grippers depends up on orientation of parts and
friction between parts and grippers.

The operations performed by the gripper are.

→ To move fingers with commanded speed.

→ To handle or grip the handled part with not
more than specified force.

→ To open and close the fingers at specified
position.

→ To measure dimensions of the parts.

Tooling: The use of grippers as end effector is limited because it may not be certain applications like welding, painting etc. A robot tool is attached directly at the end of the wrist and act as end effector.

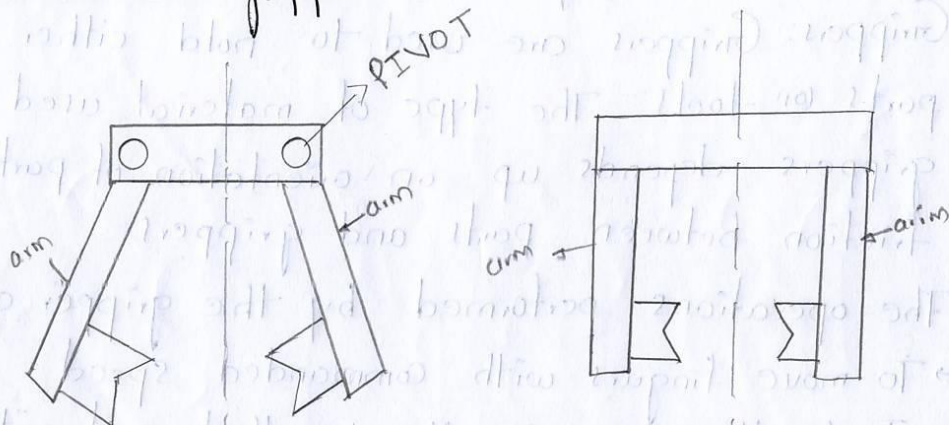
Example: Spot welding, painting, drilling etc.

Types of grippers:

The following types of grippers are.

- (1) Mechanical grippers.
- (2) Magnetised grippers.
- (3) Vacuum type grippers.
- (4) Suction type gripper.
- (5) Hooks type gripper.
- (6) Scoops
- (7) Ladders gripper.

Mechanical gripper.



Mechanical gripper is an end effector that uses mechanical fingers actuated by a mechanism to grasp an object. The fingers are attached to the mechanism.

- The function of the gripper mechanism is to translate some form of power input into the grasping action.
- The power input is supplied from the robot.
- The mechanism must be able to open & close the fingers.

Magnetic Gripper: These grippers can be divided into two types

- 1) Electromagnetic
- 2) Permanent magnetic

Electromagnetic: These grippers are easier to control but require source of DC power and an appropriate control unit. When the part is to be released the control unit reverse the polarity at reduced power level and switch of electromagnet.

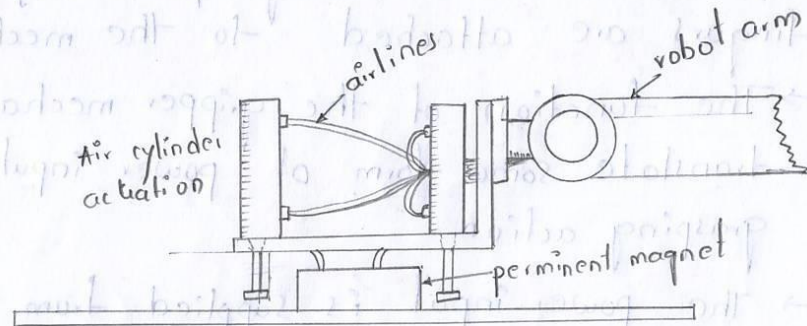
Permanent Magnetic: The permanent magnetic type of gripper when the part is to be released at the end of handling of the cycle. It means of separate of the work part from the magnet.

Advantage

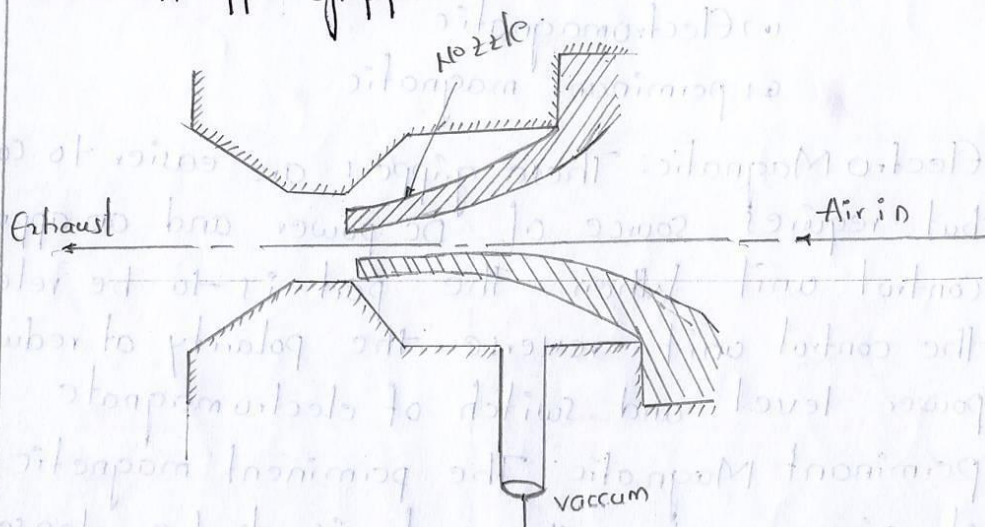
- 1) Pick up links are very fast.
- 2) They have the ability to handle metal parts.

with holes.

3) These requires only one surface for gripping.



Vacuum-type gripper:



It is used to lift the flat glass of the work. This type of grippers are made of elastic rubbers. Such as rubber, soft plastic, etc. The shape of the vacuum cup as shown. It means of removing the air between the cup and part surface to create the vacuum is

required. The vacuum pump and venturi is the¹² common devices used for those purposes.

→ The -ve air pressure between cup and the object is given by

$$F = PA$$

$$F = \text{Force}$$

$$P = \text{-ve pressure (Pressure b/w in \& out)}$$

$$A = \text{Total effective area of suction cup used to create the vacuum.}$$

A vacuum gripper is to be designed to handle flat plate glass in an automobile chilled plant each plate weight 28lb a single suction cup will be used and the diameter of suction cup is 60 inch determine the -ve pressure required to lift Each plate. Use safety factor of 1.5 inch.

$$\text{Weight} = 28 \text{ lb}$$

$$\text{Cup diameter (d)} = 60 \text{ inch}$$

$$\text{Area of Circle (A)} = \frac{\pi d^2}{4}$$

$$= \frac{\pi}{4} 60^2$$

$$= 2827.43 \text{ inch}^2$$

$$F = PA$$

$$28 = P \times 2827.43$$

$$P = \frac{28}{2827.43}$$

$$= 9.90 \times 10^{-3} \text{ lb/inch}^2$$

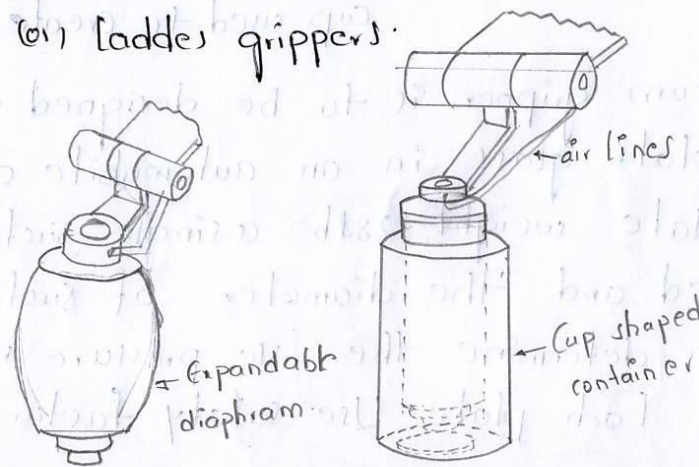
Safety factor 1.5 will be consider as a pressure

$$p = 9.90 \times 10^{-3} \times 1.5$$

$$= 14.85 \times 10^{-3} \text{ lb/inch}^2$$

Hooks gripper: Hooks can be used as end effector to handle containers of parts to load and unloading. The items to be handled by a hook must have some soft of handle to enable the hook to handle it.

Scoops (or) ladles grippers.



It can be used to handle certain materials in liquid (or) powder form and chemicals in liquid (or) powder form and food materials etc. that can be handled by a robot using this method of holding.

Constructions of mechanical gripper.

To holding the parts by friction between the fingers and the work parts. The following gripper force is given by

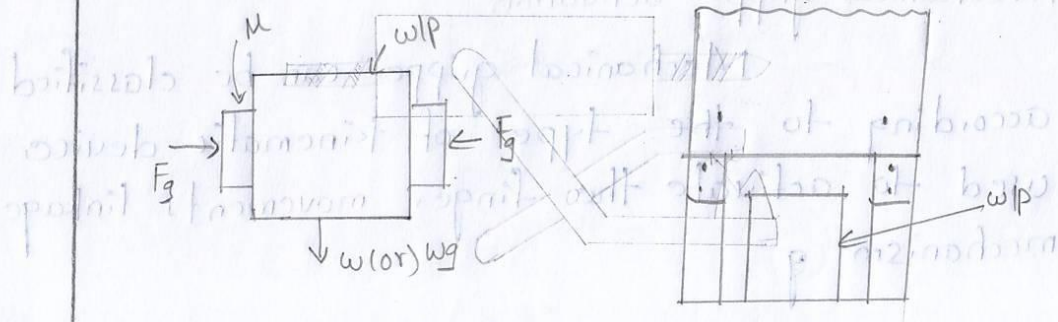


figure shows force against workpiece, parallel to finger surfaces

$$\mu \times n \times f_g = w$$

μ = Co-efficient of friction

n = No. of contacting surfaces

f = gripper force

w = Weight of workpiece.

angelburg suggested that in a high speed handling operation that would exert a force that is twice the weight of work part he reduces the problem by use of "g" factor and revised a equation on

$$\mu n f_g = w \times g$$

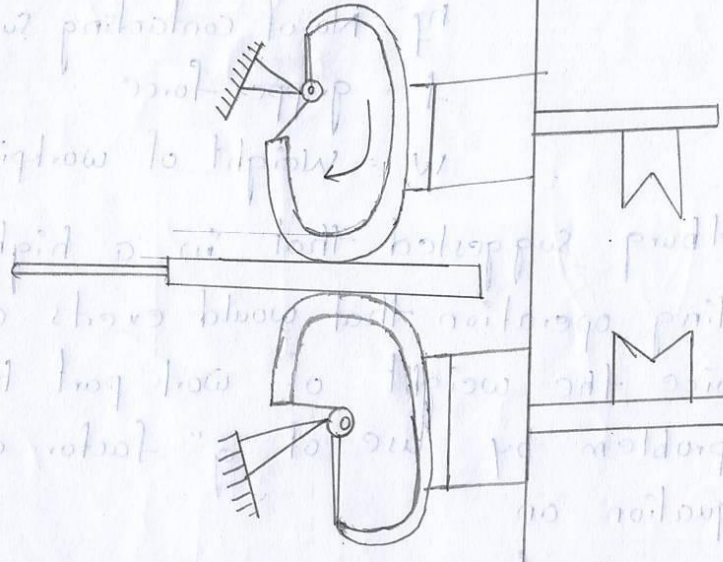
- If the acceleration force is applied in the same direction as the gravity force $g = 3.0$
- If the acceleration is applied in the opposite direction $g = 1.0$
- If the acceleration is applied in horizontal $g = 1.0$

Mechanical gripper activation.

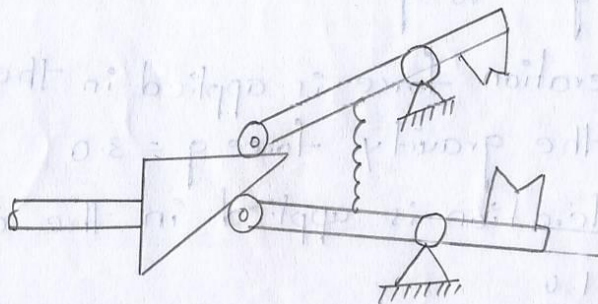
Mechanical gripper can be classified according to the type of kinematic device used to activate the finger, movement's linkage mechanism.

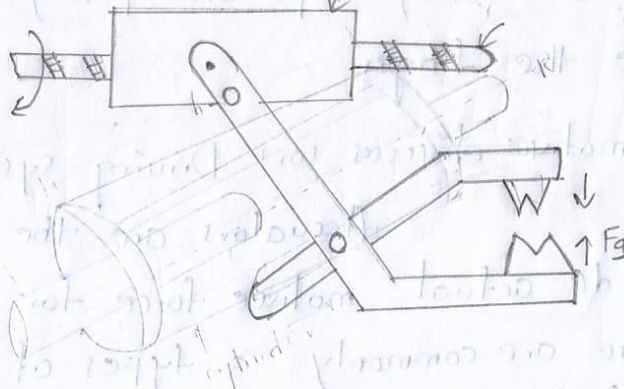
- 1) Gear and rack activation
- 2) Cam activation
- 3) Rope and pulley activation.
- 4) Screw type gripper.

(i)



(ii)



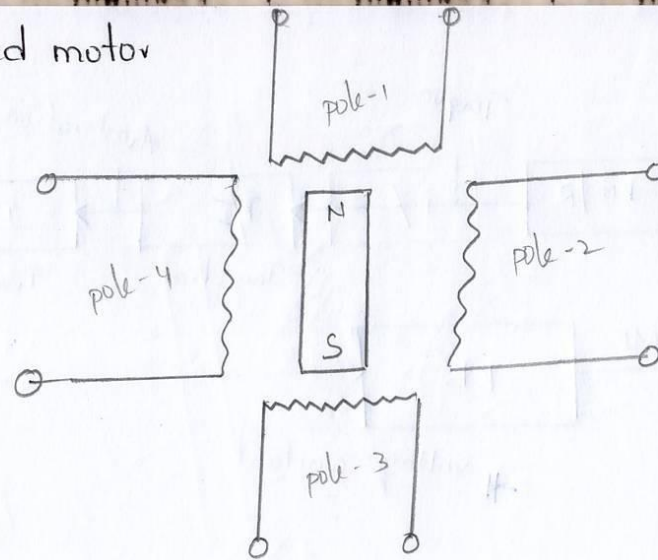


(i) The linkage actuators are available wide range of design possibilities to actuate the open and closing the fingers in this. The input power F_a to the gripper is converted into the gripping force F_g applied by the finger rack and gear actuation. The rack gear would be attached to a piston or some other mechanisms that would provide a linear motion movement of the rack would drive to partial pinion gears and there would intended open and close fingers.

Cam: A cam and follower arrangement after using a spring loaded follower can provide the open and closing the grippers.

Screw: The screw is turned by a motor, when the screw is rotated in one direction, this comes a thread and block to be translated in one direction. The screw is rotated in opposite direction, the threaded block moves in opposite direction.

Stepped motor



Speed motors are a unique type of accurate and have been used mostly in computer peripheral. In robotics stepped motors are used for relatively light duty applications and also stepped motors are typically used in open loop systems and closed loop system.

→ The relation between a stepped motor resolution and its step angle is given by

$$\eta = \frac{A}{360}$$

where

η = Resolution

A = step angle

Considerations in gripper selection and design.

- 1) The part surface must be grasped and must be reachable.
- 2) The gripper design must accommodate the change in size that occurs between part.

loading and unloading

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(3) If there is a choice between two different dimensions on the part the large dimensions should be selected for grasping

(4) Holding the part by its large surface will provide better control and stability of work part in position.